

Off-Policy Evaluation and Optimization

CML Week 7

2026-10-13

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. Balancing weights
4. Double robustness, Double ML and TMLE
5. Heterogeneous Treatment Effects
6. HTE II: Neural Networks
7. **Off-policy evaluation and optimization**
8. Experimental design
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference



From effect estimation to policy evaluation

So far in this course: Estimate treatment effects

- Average treatment effect: $\tau = \mathbb{E}[Y(1) - Y(0)]$
- Conditional average treatment effect: $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$
- Goal: understand *how large* the effect is (and for whom)

Today's shift: Evaluate and learn *policies* (decision rules)

- A policy $\pi : \mathcal{X} \to \{0, 1\}$ maps covariates to treatment assignments
- Policy value: $V(\pi) = \mathbb{E}[Y(\pi(X))]$ - expected outcome if we follow policy π
- Goal: find policies that produce good outcomes, not just estimate effects

Why this matters:

- Decision-making requires choosing who gets treated (not just knowing effect sizes)
- Can evaluate policies without estimating $\tau(x)$ for all x
- Different objective: maximize welfare vs. estimate parameters accurately



(Dudík, Langford, and Li 2011)

The contextual bandit setting

The setup: We have logged data from a *behavior policy* π_0, (X_i, A_i, Y_i) for $i = 1, \ldots, n$

- Treatments were assigned by π_0: $A_i \sim \pi_0(\cdot \mid X_i)$
  - May be stochastic: $\pi_0 : \mathcal{X} \to \Delta(\{0, 1\})$
  - Or deterministic: $\pi_0 : \mathcal{X} \to \{0, 1\}$
- Only observe outcome for chosen action: $Y_i = Y_i(A_i)$

The goal: Evaluate a *target policy* π without deploying it

- Want to estimate: $V(\pi) = \mathbb{E}_X[\mathbb{E}[Y(\pi(X)) \mid X]]$
- But data came from π_0, not π!

Overlap: $\pi_0(a \mid x) > 0$ whenever $\pi(a \mid x) > 0$

- Need positive probability of observing actions that π would take
- Without overlap: no data to evaluate π in some regions of $\mathcal{X}$

The off-policy challenge: Distribution shift between π_0 and π requires correction

Three approaches to off-policy evaluation

1. Direct Method (DM): Model the outcomes

- Estimate $\hat{\mu}(a, x) = \mathbb{E}[Y \mid A = a, X = x]$
- Predict: $\hat{V}_{DM}(\pi) = \mathbb{E}_X[\hat{\mu}(\pi(X), X)]$
- Low variance, but biased if model wrong

2. Inverse Propensity Score (IPS): Reweight the observations

- $\hat{V}_{IPS}(\pi) = \frac{1}{n} \sum_i \frac{\mathbb{1}(A_i = \pi(X_i))}{\pi_0(A_i|X_i)} Y_i$
- Unbiased if π_0 known/correct, but high variance due to the density ratio $\frac{\pi(x)}{\pi_0(x)}$.

3. Doubly Robust (DR): Combine both

- Use outcome model, correct with IPS on residuals
- Unbiased if *either* model is correct
- Best of both worlds

The doubly robust estimator

The DR formula for policy evaluation:

$$\hat{V}_{DR}(\pi) = \frac{1}{n} \sum_i \left[\hat{\mu}(\pi(X_i), X_i) + \frac{\mathbb{1}(A_i = \pi(X_i))}{\pi_0(A_i \mid X_i)} (Y_i - \hat{\mu}(A_i, X_i)) \right]$$

Intuition: Outcome model prediction + importance-weighted residual correction
Why doubly robust:

- If $\hat{\mu}(a, x)$ correct: residual term averages to zero
- If $\pi_0(a \mid x)$ correct: importance weighting fixes distribution shift
- If both correct: semiparametrically efficient

Connection to Lecture 4: Same idea, different estimand (policy value vs ATE)

From evaluation to learning

So far: Evaluate a given policy π

- Given a candidate policy, estimate its value $V(\pi)$
- Tells us if a proposed policy is good or bad

The learning problem: Find the *best* policy from data

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \hat{V}_{DR}(\pi)$$

- Optimize the DR estimate directly over policy class Π
- Can use gradient-based methods (if π parameterized) or tree-based methods

Connection to HTE estimation:

- If we knew $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ perfectly: $\pi^*(x) = \mathbb{1}(\tau(x) > 0)$
- But we don't need to estimate $\tau(x)$ everywhere to learn a good policy
- Only need to get the *sign* right at observed covariate values
- Policy learning can succeed even when effect estimation is hard



Practical considerations

Strengths of the DR approach:

- **Flexibility:** Works with any ML methods for $\hat{\mu}$ and π_0
- **Robustness:** Two chances to get it right (either model sufficient)
- **Foundation:** Underlies modern off-policy evaluation and bandit algorithms

Limitations:

- **Overlap requirement:** Breaks down when $\pi_0(a \mid x) \approx 0$ but $\pi(a \mid x) > 0$
- **Model selection:** Still need to choose and validate both models
- **Finite sample:** Theoretical guarantees are asymptotic

Takeaways

- Policy evaluation ≠ effect estimation (different objectives)
- Can learn from logged/observational data without deploying new policies
- DR provides insurance against misspecification

Open questions

1. How do we learn optimal policies with statistical guarantees?
   - *Motivates:* (Athey and Wager 2021)
2. When should we optimize decisions vs. estimate effects?
   - *Motivates:* (Fernández-Loría and Provost 2022)
3. How do we *superlearn* an optimal treatment rule?
   - *Motivates:* (Luedtke and van der Laan 2016)
4. When overlap is poor, the importance weights explode. Can we shrink them to trade a little bias for much less variance?
   - *Motivates:* (Su et al. 2020)

More to read

Paper	Why you'd read it
(Athey and Wager 2021)	Learning optimal policies from observational data.
(Fernández-Loría and Provost 2022)	Decision-making versus effect estimation objectives.
(Luedtke and van der Laan 2016)	Data-adaptive rules for dynamic treatment (TMLE).
(Su et al. 2020)	Shrinking importance weights in DR estimators to cut variance.
(Kern et al. 2025)	Why causal reasoning matters for algorithms.
(Swaminathan and Joachims 2015)	Learning policies from logged bandit data.
(Jiang and Li 2016)	Doubly robust methods for value estimation.
(Kallus 2018)	Covariate balancing for policy evaluation.
(Kitagawa and Tetenov 2018)	Welfare-based optimal treatment assignment rules.

References

Athey, Susan, and Stefan Wager. 2021. "Policy Learning with Observational Data." *Econometrica* 89 (1): 133–61.

Dudík, Miroslav, John Langford, and Lihong Li. 2011. "Doubly Robust Policy Evaluation and Learning." In *International Conference on Machine Learning*, 1097–1104. PMLR.

Fernández-Loría, Carlos, and Foster Provost. 2022. "Causal Decision Making and Causal Effect Estimation Are Not the Same...and Why It Matters." *INFORMS Journal on Data Science* 1 (1): 4–16.

Jiang, Nan, and Lihong Li. 2016. "Doubly Robust Off-Policy Value Evaluation." In *International Conference on Machine Learning*, 652–61. PMLR.

Kallus, Nathan. 2018. "Balanced Policy Evaluation and Learning." *Advances in Neural Information Processing Systems* 31.

Kern, Christoph, Urs Fischer-Abaigar, Jochen Schweisthal, et al. 2025. "Algorithms for Reliable Decision-Making Need Causal Reasoning." *Nature Computational Science* 5: 356–60. https://doi.org/10.1038/s43588-025-00814-9.

Kitagawa, Toru, and Aleksey Tetenov. 2018. "Who Should Be Treated? Empirical Welfare Maximization Methods for Treatment Choice." *Econometrica* 86 (2): 591–616.

Luedtke, Alexander R, and Mark J van der Laan. 2016. "Super-Learning of an Optimal Dynamic Treatment Rule." *International Journal of Biostatistics* 12 (1): 305–32.

Su, Yi, Maria Dimakopoulou, Akshay Krishnamurthy, and Miroslav Dudík. 2020. "Doubly Robust Off-Policy Evaluation with Shrinkage." In *Proceedings of the 37th International Conference on Machine Learning*, 119:9167–76. Proceedings of Machine Learning Research. PMLR.

Swaminathan, Adith, and Thorsten Joachims. 2015. "Counterfactual Risk Minimization: Learning from Logged Bandit Feedback." In *International Conference on Machine Learning*, 814–23. PMLR.